

Assignment # 2

MAT 123

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Due date: 10th May, 2026

Problems:

1. Question 1: Let $f(x)$ be a function that is defined and has a continuous derivative on the interval $(2, \infty)$. Assume also that $f(3) = -12$, $|f(x)| < x^5 + 8$ and $\int_3^\infty f(x)e^{-x/4}dx = -9$. Determine the value of $\int_3^\infty f'(x)e^{-x/4}dx$?

2. Find $f'(2)$ if $f(x) = e^{g(x)}$ and $g(x) = \int_2^x \frac{t}{1+t^4}dt$.

3. Show that if p is a positive integer, then the power series

$$\sum_{k=0}^{\infty} \frac{(pk)!}{(k!)^p} x^k$$

has a radius of convergence of $\frac{1}{p^p}$.